Grok Bots Case Study

Multi-Agent Orchestration for Operational Workflows

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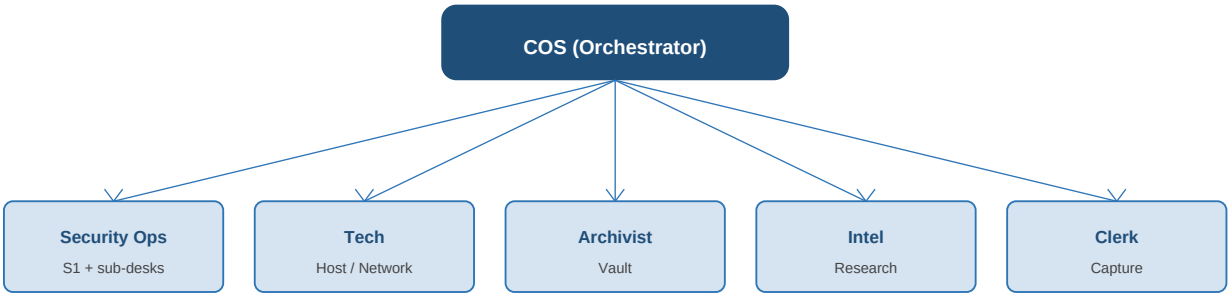
System Hierarchy



Overview

A hierarchical multi-agent system that turns recurring operational work into reliable, low-intervention processes. A central orchestrator (Chief of Staff) coordinates specialised agents. Each agent owns a clear domain, returns structured status with file paths, and stops at explicit approval gates for consequential actions.

Specialised Desks

Desk	Purpose
Security Ops (S1)	Wazuh agent health, detection rules, alert correlation, visual monitoring
Tech (S4)	Host & network baselines, drift detection, scheduled health sweeps, runbooks
Archivist (S10)	Obsidian vault capture, formatting, filing, and index maintenance
Intel (S2)	Structured research briefs on named subjects
Clerk (S9)	Operational document capture and allocation against defined rules

Key Mechanisms

Skills — reusable instruction sets (steps, decision rules, expected outputs, safety boundaries). **Routines** — scheduled or event-driven execution of skills. **Approval gates** — consequential actions stop at “AWAITING APPROVAL” with exact next steps. **Status reporting** — short status line (done / blocked / in progress) plus file path. **Boundaries** — written purpose and explicit limits per agent.

Worked Example — Daily Host & Network Health Sweep (Tech desk)

Trigger: Scheduled routine (daily).
What it checks: Host uptime and load, disk and memory thresholds, key service status, interface state, and drift against the last known baseline. Flags items outside expected ranges.
Output: Structured status written to a dated file under the Tech desk workspace. Non-consequential findings are logged. Anything that would change configuration or restart a service is held.
Approval gate: Proposed remediation steps are listed and the routine ends with **AWAITING APPROVAL**. No changes are applied until the orchestrator (or human) explicitly releases them.
Sample status returned to COS:

```
Tech — daily health sweep — done. 3 hosts within baseline; 1 host disk at 87% (threshold 80%). Proposed: clean /var/log on host-03. AWAITING APPROVAL. Report: artifacts/tech/health-2026-10-25.md
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What This Demonstrates

Practical multi-agent design, reusable workflow patterns, human-in-the-loop controls, and operational reporting. The same principles apply to embedding AI tools and agents into software delivery teams: identify recurring work, decompose it into specialised roles, add clear boundaries and approval points, and iterate from real use.

Further evidence available on request: additional agent configuration excerpts, skill and routine definitions, sample status reports, and a short live walkthrough.